

MOMBETSU, Japan

WMO: 474350

Lat: 44.3452N

Lon: 143.3560E

Elev: 18

StdP: 101.11

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.4	-11.7	-17.8	0.8	-12.3	-16.3	0.9	-10.5	10.9	-3.3	9.7	-3.9	3.2	210	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.5	27.8	20.8	25.7	20.1	23.9	19.5	22.5	26.2	21.5	24.4	20.5	23.0	3.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	15.9	24.6	20.3	15.0	23.6	19.4	14.2	22.5	66.8	26.4	62.9	24.5	59.3	23.1	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.0	6.9	-16.0	32.4	2.0	2.1	-17.5	33.9	-18.6	35.1	-19.8	36.3	-21.2	37.9		
			-16.5	24.2	2.0	1.1	-17.9	25.0	-19.0	25.7	-20.1	26.3	-21.5	27.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-5.6	-5.7	-1.4	4.7	9.8	13.1	17.3	19.7	16.8	10.5	3.5	-2.8	(6)
(7)		DBStd	9.48	3.04	3.60	3.66	4.13	4.71	3.82	3.64	3.15	2.96	3.20	3.98	3.13	(7)
(8)		HDD10.0	2146	484	439	353	169	63	12	0	0	0	33	197	396	(8)
(9)		HDD18.3	4357	743	672	611	409	267	162	65	21	64	243	444	654	(9)
(10)		CDD10.0	953	0	0	0	9	57	106	227	299	204	48	3	0	(10)
(11)		CDD18.3	121	0	0	0	0	2	5	33	62	18	0	0	0	(11)
(12)		CDH23.3	523	0	0	0	3	27	32	136	258	67	1	0	0	(12)
(13)		CDH26.7	117	0	0	0	1	6	5	35	62	9	0	0	0	(13)
(14)	Wind	WSAvg	3.1	3.6	3.4	3.5	3.4	2.9	2.4	2.1	2.3	3.0	3.4	3.5	3.6	(14)
(15)	Precipitation	PrecAvg	855	44	34	36	48	60	65	110	119	136	75	65	63	(15)
(16)		PrecMax	1132	121	71	71	150	142	131	248	318	313	159	155	147	(16)
(17)		PrecMin	540	21	12	11	7	18	22	18	18	54	21	19	28	(17)
(18)		PrecStd	155	22	16	16	29	31	30	54	76	79	33	35	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	6.7	11.9	22.0	26.5	26.9	30.7	31.5	27.9	21.6	16.7	8.3	(19)
(20)			MCWB	1.5	3.2	6.1	11.5	14.8	16.4	22.5	23.2	20.1	14.9	10.0	5.2	(20)
(21)		2%	DB	1.2	3.5	8.0	17.0	22.6	23.3	26.8	28.3	25.2	19.2	13.7	5.3	(21)
(22)			MCWB	-0.7	0.4	3.6	9.2	13.3	16.7	21.1	22.1	18.9	13.5	8.9	2.7	(22)
(23)		5%	DB	0.0	1.3	5.7	14.0	19.6	20.8	24.3	26.0	23.2	17.3	11.5	3.3	(23)
(24)			MCWB	-1.3	-1.1	2.1	7.8	12.6	15.9	20.5	21.4	18.2	12.9	7.7	1.2	(24)
(25)		10%	DB	-1.0	-0.5	3.8	11.3	17.0	18.8	22.5	24.2	21.6	15.8	9.5	2.0	(25)
(26)			MCWB	-2.4	-2.2	0.9	6.3	11.5	15.0	19.6	20.7	17.5	11.9	6.6	0.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	3.1	6.5	12.3	16.1	19.1	23.7	24.1	22.0	16.4	11.5	5.6	(27)
(28)			MCDWB	4.3	6.7	11.5	19.1	23.5	24.3	27.7	29.4	26.0	19.3	15.3	7.8	(28)
(29)		2%	WB	-0.1	0.6	4.0	9.9	14.5	17.5	22.2	22.9	20.4	14.8	9.5	3.0	(29)
(30)			MCDWB	0.9	3.2	7.3	16.5	20.7	21.7	25.7	26.8	23.6	17.7	12.7	4.7	(30)
(31)		5%	WB	-1.2	-0.9	2.5	8.1	13.2	16.4	20.9	22.0	19.2	13.6	8.1	1.5	(31)
(32)			MCDWB	-0.1	0.8	5.3	13.2	18.7	20.0	23.6	25.1	21.8	16.4	11.1	3.1	(32)
(33)		10%	WB	-2.4	-2.0	1.1	6.6	11.9	15.3	19.7	21.1	18.1	12.4	6.8	0.2	(33)
(34)			MCDWB	-1.1	-0.6	3.5	11.0	16.2	18.3	22.2	23.7	20.7	15.3	9.4	1.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	6.6	6.5	7.4	7.5	6.1	5.4	5.5	7.2	7.9	6.4	5.7	(35)
(36)			MCDBR	7.0	9.4	9.9	12.7	13.0	10.0	8.9	8.6	9.2	9.9	10.2	7.6	(36)
(37)			MCWBR	6.0	7.2	6.9	7.1	6.3	5.1	4.5	4.2	5.2	6.3	6.9	6.2	(37)
(38)		5% WB	MCDWR	6.6	8.7	9.6	11.7	11.4	8.8	8.1	7.4	7.6	8.7	9.5	7.3	(38)
(39)			MCWBR	5.9	6.9	6.9	7.1	6.1	4.9	4.6	3.9	4.8	6.1	6.8	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.373	0.437	0.496	0.513	0.495	0.477	0.454	0.403	0.396	0.372	0.346	(40)	
(41)		taud	2.071	1.892	1.827	1.853	1.916	2.065	2.184	2.266	2.358	2.227	2.185	2.093	(41)	
(42)		Ebn at Noon	748	774	775	768	770	787	797	802	818	758	699	681	(42)	
(43)		Edh at Noon	103	149	182	193	187	162	142	127	107	106	93	92	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.42	2.36	3.50	4.46	4.82	4.85	4.57	4.21	3.57	2.57	1.51	1.14	(44)	
(45)		RadStd	0.12	0.26	0.28	0.35	0.40	0.47	0.55	0.27	0.30	0.13	0.13	0.09	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page